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Ingé 4 Cyber sécurité Groupe 01

# Mini-projet 2 : Contrôle d'Accès et Cryptographie

**04/02/2026**

1. Création d'Utilisateurs et Groupes (si nécessaire):

```
┌──(kali㉿kali)-[~]
└─$ sudo adduser alice
[sudo] password for kali:
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for alice
Enter the new value, or press ENTER for the default
        Full Name []:
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] y
┌──(kali㉿kali)-[~]
└─$ sudo adduser bob
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for bob
Enter the new value, or press ENTER for the default
        Full Name []:
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] y

┌──(kali㉿kali)-[~]
└─$ sudo addgroup dev
┌──(kali㉿kali)-[~]
└─$ sudo usermod -aG dev alice
```

2. Création de Fichiers et Répertoires de Test :

```
┌──(kali㉿kali)-[~]
└─$ mkdir test_dir

┌──(kali㉿kali)-[~]
└─$ touch test_file.txt

┌──(kali㉿kali)-[~]
└─$ echo "Contenu de test" > test_file.txt

┌──(kali㉿kali)-[~]
└─$
```

3. Affichage des Permissions Actuelles:

```
┌──(kali㉿kali)-[~]
└─$ ls -l test_file.txt
-rw-rw-r-- 1 kali kali 16 Feb  6 09:29 test_file.txt

┌──(kali㉿kali)-[~]
└─$ ls -l test_dir
total 0
```

4. Modification des Permissions avec chmod:

-On donne la propriété du fichier à alice

```
┌──(kali㉿kali)-[~]
└─$ sudo chown alice test_file.txt
[sudo] password for kali:

┌──(kali㉿kali)-[~]
└─$ sudo chmod 600 test_file.txt

┌──(kali㉿kali)-[~]
└─$ ls -l test_file.txt
-rw------- 1 alice kali 16 Feb  6 09:29 test_file.txt
```

5. Changement de Propriétaire et de Groupe avec chown et chgrp

Objectif : Groupe dev = Read/Write/Execute.

```
┌──(kali㉿kali)-[~]
└─$ sudo chgrp dev test_dir
[sudo] password for kali:

┌──(kali㉿kali)-[~]
└─$ sudo chmod g+rwx test_dir
```

6. Vérifications

```
┌──(kali㉿kali)-[~]
└─$ sudo -u bob cat test_file.txt
cat: test_file.txt: Permission denied

┌──(kali㉿kali)-[~]
└─$ sudo -u alice touch test_dir/test_alice.txt
touch: cannot touch 'test_dir/test_alice.txt': Permission denied

┌──(kali㉿kali)-[~]
└─$
```

***Partie 2B : Chiffrement Symétrique avec OpenSSL***

créez le fichier secret

```
┌──(kali㉿kali)-[~]
└─$ echo "Ceci est un texte confidentiel pour le projet ECE." > secret.txt
```

Chiffrement du fichier

```
┌──(kali㉿kali)-[~]
└─$ openssl enc -aes-256-cbc -salt -in secret.txt -out secret.txt.enc -pbkdf2
enter AES-256-CBC encryption password:
Verifying - enter AES-256-CBC encryption password:
```

Vérification

```
┌──(kali㉿kali)-[~]
└─$ cat secret.txt.enc
Salted__◆й◆7◆l◆&9-p◆◆◆3s{◆◆◆◆◆~◆◆◆◆7-◆◆&◆◆<◆◆e◆◆'
◆◆^3◆◆◆S_&◆3 }◆◆3
```

Déchiffrement du fichier

```
┌──(kali㉿kali)-[~]
└─$ openssl enc -aes-256-cbc -d -in secret.txt.enc -out secret_decypted.txt -pbkdf2
enter AES-256-CBC decryption password:
```

```
┌──(kali㉿kali)-[~]
└─$ openssl enc -aes-256-cbc -d -in secret.txt.enc -out secret_decypted.txt -pbkdf2
enter AES-256-CBC decryption password:

┌──(kali㉿kali)-[~]
└─$ cat secret_decypted.txt
Ceci est un texte confidentiel pour le projet ECE.

┌──(kali㉿kali)-[~]
```

***Création et manipulation de certificats auto-signés***

Génération de la Clé Privée

```
┌──(kali㉿kali)-[~]
└─$ openssl genrsa -out cle_privee.pem 2048
```

Création de la CSR (Certificate Signing Request)

```
┌──(kali㉿kali)-[~]
└─$ openssl req -new -key cle_privee.pem -out requete.csr
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:fr
State or Province Name (full name) [Some-State]:Paris
Locality Name (eg, city) []:paris
Organization Name (eg, company) [Internet Widgits Pty Ltd]:ece
Organizational Unit Name (eg, section) []:
Common Name (e.g. server FQDN or YOUR name) []:mimene
Email Address []:imene.maouche@edu.ece.fr

Please enter the following 'extra' attributes
to be sent with your certificate request
A challenge password []:
An optional company name []:

┌──(kali㉿kali)-[~]
└─$
```

Signature du Certificat (Auto-signature)

```
┌──(kali㉿kali)-[~]
└─$ openssl x509 -req -in requete.csr -signkey cle_privee.pem -out certificat.crt -days 365
Certificate request self-signature ok
subject=C=fr, ST=Paris, L=paris, O=ece, CN=mimene, emailAddress=imene.maouche@edu.ece.fr
```

Vérification et Manipulation

```
┌──(kali㉿kali)-[~]
└─$ openssl x509 -in certificat.crt -text -noout
Certificate:
    Data:
        Version: 3 (0x2)
        Serial Number:
            2b:16:12:50:87:42:88:5e:1e:52:3d:83:60:6b:c0:ad:4e:9e:d5:75
        Signature Algorithm: sha256WithRSAEncryption
        Issuer: C=fr, ST=Paris, L=paris, O=ece, CN=mimene, emailAddress=imene.maouche@edu.ece.fr
        Validity
            Not Before: Feb  7 23:01:22 2026 GMT
            Not After : Feb  7 23:01:22 2027 GMT
        Subject: C=fr, ST=Paris, L=paris, O=ece, CN=mimene, emailAddress=imene.maouche@edu.ece.fr
        Subject Public Key Info:
            Public Key Algorithm: rsaEncryption
                Public-Key: (2048 bit)
                Modulus:
                    00:98:2e:7e:e5:dd:20:a9:2a:e8:ea:f7:10:88:73:
                    6b:f2:a9:4e:6a:eb:02:b4:bb:82:24:ed:ae:81:c5:
                    a8:66:2a:ac:d8:4a:fd:87:aa:b8:9f:f4:84:a5:13:
                    ec:89:10:d7:d8:90:7f:b2:86:07:39:8b:0e:79:22:
                    eb:5c:6a:79:df:2c:d9:3f:80:7d:62:05:0d:b5:55:
                    96:80:52:d2:19:84:93:90:aa:8b:d9:51:ad:42:ea:
                    69:50:12:2d:ff:98:3a:05:e9:9f:c0:43:cc:4a:92:
                    ad:87:49:2b:8c:b5:50:24:b3:7a:fb:05:be:27:8d:
                    4d:23:a0:d8:74:13:b4:b8:e1:78:b2:4b:0b:67:22:
                    42:1b:1d:c4:c3:d6:cc:c3:87:88:ac:1f:56:5a:0d:
                    bf:cb:21:10:81:1f:01:15:34:50:d0:7a:b0:c5:12:
                    7b:fc:48:dc:dc:62:a3:35:be:9d:94:62:c7:d4:35:
                    d2:e8:e0:00:f7:76:b3:16:d9:67:3c:13:5d:2f:70:
                    f8:52:50:42:da:99:f2:28:80:c7:cf:c5:d0:7f:9a:
                    ab:e7:82:d2:92:c0:20:0d:f6:c7:7a:8d:2d:35:f6:
                    02:79:6a:a5:37:7e:78:32:3e:7f:b7:24:02:9f:e1:
                    c6:ad:5f:ce:ff:d5:c6:0d:69:37:a3:a9:d7:14:36:
                    bf:d9
                Exponent: 65537 (0x10001)
        X509v3 extensions:
            X509v3 Subject Key Identifier:
                88:BA:47:4F:B3:D7:4C:2B:47:72:43:1A:D5:0E:7D:3C:D2:97:70:C5
    Signature Algorithm: sha256WithRSAEncryption
    Signature Value:
```

Pour extraire la **clé publique**

```
┌──(kali㉿kali)-[~]
└─$ openssl x509 -in certificat.crt -pubkey -noout > cle_publique.pem

┌──(kali㉿kali)-[~]
```

***Partie 2D : Mise en place d'un VPN (OpenVPN)***

Préparation (Easy-RSA)

```
┌──(root㉿kali)-[/home/kali]
└─# cd /etc/openvpn/easy-rsa

┌──(root㉿kali)-[/etc/openvpn/easy-rsa]
└─# ./easyrsa build-ca nopass
Using Easy-RSA 'vars' configuration:
* /etc/openvpn/easy-rsa/vars

EasyRSA version 3.2.2

Error
------
EASYRSA_PKI does not exist (perhaps you need to run init-pki)?
Expected to find the EASYRSA_PKI at:
* /etc/openvpn/easy-rsa/pki

Run easyrsa without commands for usage and command help.


┌──(root㉿kali)-[/etc/openvpn/easy-rsa]
└─# ./easyrsa init-pki
Using Easy-RSA 'vars' configuration:
* /etc/openvpn/easy-rsa/vars

Notice
------
'init-pki' complete; you may now create a CA or requests.

Your newly created PKI dir is:
* /etc/openvpn/easy-rsa/pki
```

## Créer la CA (Autorité de Certification)

```
┌──(root㉿kali)-[/etc/openvpn/easy-rsa]
└─# ./easyrsa build-ca nopass
Using Easy-RSA 'vars' configuration:
* /etc/openvpn/easy-rsa/vars
.+.............+......+....+ ... +.........+.....+.++++++++++++++++++++++++++++++++++++++++* .. +.+ .. + ... ++++++++++++++++++++++++++++++++++
+++++* .. +.+ ... + ... +.....+ ... +.......+.....+.........................+.+ .. +....+......+ .. +.+.......+ ... + ... +........+......+.....+ ..
.. + ... +.....+ ... +.......+.............+.......+.....+ .. + ... +.........+.....+.................+.....+............
+....+.....+......+.......+.......+......+................+.....+ ... +.......+................+......+ ... + .. + ... +......+....+ .. +
.................+.+ .. +.......+.....++++++
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++++++++++++++++++++*....+................+ ... +....+ ... + ... +.....+......+.+ .. +.......+.............+.......+.....+.......+ ... +....
....+.........................+ .. +....+......+ ... + .. + ... +.......+ ... +.......+......+ ... + .. +....+.....+ ... + ... + ... +....+......+.....+ ..
.. + .. +....+ .. +.......+......+.........+.........+......+.+.........+......+ .. +............+.....+....+......+ ... +.....+...........
.+....+.....+....+.....+....+ ... +................+.............+ ... + ... +.+.....+......+.........+.+.....+ ... +.......+ ... +.........
+ .. + ... +.+.....+.+........+.......+ ... +..........+.+.........+.....+.....+ ... +.+...........++++++
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Common Name (eg: your user, host, or server name) [Easy-RSA CA]:mimene

Notice
------
CA creation complete. Your new CA certificate is at:
* /etc/openvpn/easy-rsa/pki/ca.crt

Create an OpenVPN TLS-AUTH|TLS-CRYPT-V1 key now: See 'help gen-tls'

Build-ca completed successfully.

┌──(root㉿kali)-[/etc/openvpn/easy-rsa]
```

## Générer le certificat du serveur

```
┌──(root㉿kali)-[/etc/openvpn/easy-rsa]
└─# ./easyrsa build-server-full server nopass
Using Easy-RSA 'vars' configuration:
* /etc/openvpn/easy-rsa/vars
..............+.........+......+........+....+ ... +.....+ ... +..........+ .. +.........+ ... +.......+ ... +.....+......+.......+ ... +......+
 ... + .. + ... ++++++++++++++++++++++++++++++++++++++++* ... + ... +.+ ... ++++++++++++++++++++++++++++++++++++++++*..............+.........+....
 .. +......+ ... +......+.+.....+.+.........++++++
................+.......+ .. + ... +.......+......+ .. + ... + ... +..........++++++++++++++++++++++++++++++++++++++++*..........+.....+.+ ... +.
....+......+.+ .. +.++++++++++++++++++++++++++++++++++++++++*................+ ... + .. +.........+.......+ ... + .. +....+ ... +......+.....+
......+.+.........+.........................+.........+.+ .. +.........+.....+......+.+.......+ ... +..........+................+ .. +.+.
.....+.....+......+....+.....+ ... + .. +.......+ .. + ... +.........+.....+.....+.........+ ... + ... +.........+.......+.....+.....+....
...........+......+.+.....+ ... +.......+ .. +.......+.........+......+ ... + ... +.........+.....+.+..............+.....+.......+..........
 .. + ... +........+ ... +........+.++++++
 ----

Notice
------
Private-Key and Public-Certificate-Request files created.
Your files are:
* req: /etc/openvpn/easy-rsa/pki/reqs/server.req
* key: /etc/openvpn/easy-rsa/pki/private/server.key

You are about to sign the following certificate:

  Requested CN:     'server'
  Requested type:   'server'
  Valid for:        '825' days


subject=
    commonName                = server

Type the word 'yes' to continue, or any other input to abort.
  Confirm requested details: yes

Using configuration from /etc/openvpn/easy-rsa/pki/844ff591/temp.6.1
Check that the request matches the signature
Signature ok
The Subject's Distinguished Name is as follows
commonName            :ASN.1 12:'server'
Certificate is to be certified until May 12 23:12:00 2028 GMT (825 days)
```

Générer les paramètres Diffie-Hellman

```
┌──(root㉿kali)-[/etc/openvpn/easy-rsa]
└─# openssl dhparam -out /etc/openvpn/server/dh.pem 2048
Generating DH parameters, 2048 bit long safe prime
.....+.++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++*
```

Le transfert des fichiers

copier les fichiers générés vers le dossier de configuration d'OpenVPN pour que le service puisse les lire

```
┌──(root㉿kali)-[/etc/openvpn/easy-rsa]
└─# cp pki/ca.crt /etc/openvpn/server/

┌──(root㉿kali)-[/etc/openvpn/easy-rsa]
└─# cp pki/issued/server.crt /etc/openvpn/server/

┌──(root㉿kali)-[/etc/openvpn/easy-rsa]
└─# cp pki/private/server.key /etc/openvpn/server/

┌──(root㉿kali)-[/etc/openvpn/easy-rsa]
└─#
```

Préparer le fichier de configuration serveur

```
# configuration example.
ca ca.crt
cert server.crt
key server.key  # This file should be kept secret

# Diffie hellman parameters.
# Generate your own with:
#   openssl dhparam -out dh2048.pem 2048
dh dh.pem

# Allow to connect to really old OpenVPN versions
# without AEAD support (OpenVPN 2.3.x or older)
# This adds AES-256-CBC as fallback cipher and
# keeps the modern ciphers as well.
```

Génération des clés du Client

```
┌──(root㉿kali)-[/etc/openvpn/easy-rsa]
└─# cd /etc/openvpn/easy-rsa

┌──(root㉿kali)-[/etc/openvpn/easy-rsa]
└─# ./easyrsa build-client-full client1 nopass
Using Easy-RSA 'vars' configuration:
* /etc/openvpn/easy-rsa/vars
...+.+.................+........+...+......+...+.+.........+..+...+.+......+...++++++++++++++++++++++++++++++++++++++++*.......+...++
++++++++++++++++++++++++++++++++++++++++*...+.........+.........+......+.+.........+............+..+.+..+....+...+.................+...
+...........++++++
.....+...+.++++++++++++++++++++++++++++++++++++++++*..+...+..........+...+.........+...+.....+............+.+.........+.....+.+.......
...........++++++++++++++++++++++++++++++++++++++++*.....+........+...+.+...+............+.+........+.......+......+........+......+...
...+.......+..+....+..+.........+...+..........+.....+..........++++++
-----

Notice
------

Private-Key and Public-Certificate-Request files created.
Your files are:
* req: /etc/openvpn/easy-rsa/pki/reqs/client1.req
* key: /etc/openvpn/easy-rsa/pki/private/client1.key

You are about to sign the following certificate:

  Requested CN:     'client1'
  Requested type:   'client'
  Valid for:        '825' days


subject=
    commonName                = client1

Type the word 'yes' to continue, or any other input to abort.
  Confirm requested details: yes

Using configuration from /etc/openvpn/easy-rsa/pki/b65a2657/temp.6.1
Check that the request matches the signature
Signature ok
```

Assemblage du fichier client.ovpn

```
Write out database with 1 new entries
Database updated

Notice
------

Inline file created:
* /etc/openvpn/easy-rsa/pki/inline/private/client1.inline


Notice
------

Certificate created at:
* /etc/openvpn/easy-rsa/pki/issued/client1.crt


┌──(root㉿kali)-[/etc/openvpn/easy-rsa]
└─# touch client1.ovpn
```

```
┌──(root㉿kali)-[/etc/openvpn/easy-rsa]
└─# cat << EOF > client1.ovpn
client
dev tun
proto udp
remote 192.168.170.128 1194
resolv-retry infinite
nobind
persist-key
persist-tun
remote-cert-tls server
cipher AES-256-GCM
verb 3

<ca>
$(cat /etc/openvpn/server/ca.crt)
</ca>
<cert>
$(cat pki/issued/client1.crt)
</cert>
<key>
$(cat pki/private/client1.key)
</key>
EOF
```

```
┌──(root㉿kali)-[/etc/openvpn/easy-rsa]
└─# cat client1.ovpn
client
dev tun
proto udp
remote 192.168.170.128 1194
resolv-retry infinite
nobind
persist-key
persist-tun
remote-cert-tls server
cipher AES-256-GCM
verb 3

<ca>
-----BEGIN CERTIFICATE-----
MIIDPDCCAiSgAwIBAgIUWt1gpKSbY0qMJbpV8guatH+F3OowDQYJKoZIhvcNAQEL
BQAwETEPMA0GA1UEAwwGbWltZW5lMB4XDTI2MDIwNzIzMTAwOFoXDTM2MDIwNTIz
MTAwOFowETEPMA0GA1UEAwwGbWltZW5lMIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8A
MIIBCgKCAQEAm66bP5grudfoVzBBz3W4BSnEGGhqZhO7II9E55Y4b+BkJe0OVD1W
D9j2qTjlFvnAnsXOKn0lpIVo9AqV41jCC7X7nKG7ycwusEbbwEr0c28EllF2OYS2
V6REZhG/EgPUOKylrk0Ge9sosUP/6FvfYuBqwScphF+zLVp3A/idLPszbkoX9dch
ADtSdfEL+wsmnmV8mKDBWfsMqGR8Hc6g0lIiNNLiPL8ip+qPVM+12mPQtXT4W9bl
egmarBvzK5fi+XP3xem0F562mtQl8yAIVUw3VL/IcrotlOIx/ljFaXqQzOkjRYW3
B0iXj9vMErFWkTDTyrSchjXThrRxhKxEiQIDAQABo4GLMIGIMAwGA1UdEwQFMAMB
Af8wHQYDVR0OBBYEFHGNV1SHFcLjJ1RkCHdFsBzva+4PMEwGA1UdIwRFMEOAFHGN
V1SHFcLjJ1RkCHdFsBzva+4PoRWkEzARMQ8wDQYDVQQDDAZtaW1lbmWCFFrdYKSk
m2NKjCW6VfILmrR/hdzqMAsGA1UdDwQEAwIBBjANBgkqhkiG9w0BAQsFAAOCAQEA
Si56uOUGy9V86l+cuw1mpa461nSgpFSGDVXkz6PsxV4JeTieIswiYUOWCdYrDFNW
4m1Y4rslRBfy/YDWGtxIfosnJOoQURkki9sXX7AQuTeqqZs0Ygmt1qgiNOL2lz6U
IWVk7LbLC03SKi3A9ogDp2uPm5RervqLuc1b7gOqLB6kMNGOBcMtW3NR6p00dTLB
LsdPn61qsAccTAgz5ihpIvPl8L1u0OuRgardXZtVCyAZjRPXJxRaR1urRq0q7tYj
A1nD9W48ez2NAEJIlEUn8lAj30t1KCvNfaXbGmgvLVo4qLR8CZM1iYKBJxzstzor
n746rbZTKFtJBP4l0FccUg==
-----END CERTIFICATE-----
</ca>
<cert>
Certificate:
    Data:
        Version: 3 (0×2)
        Serial Number:
```

```
┌──(root㉿kali)-[/etc/openvpn/easy-rsa]
└─# nano /etc/openvpn/server.conf

┌──(root㉿kali)-[/etc/openvpn/easy-rsa]
└─# ls -l /etc/openvpn/ca.crt /etc/openvpn/server.crt /etc/openvpn/server.key /etc/openvpn/dh.pem
-rw------- 1 root root 1184 Feb  7 18:36 /etc/openvpn/ca.crt
-rw-r--r-- 1 root root  428 Feb  7 18:36 /etc/openvpn/dh.pem
-rw------- 1 root root 4485 Feb  7 18:36 /etc/openvpn/server.crt
-rw------- 1 root root 1708 Feb  7 18:36 /etc/openvpn/server.key

┌──(root㉿kali)-[/etc/openvpn/easy-rsa]
└─# openvpn --config /etc/openvpn/server.conf
2026-02-07 18:39:03 Note: --cipher is not set. OpenVPN versions before 2.5 defaulted to BF-CBC as fallback when cipher negotiation fa
iled in this case. If you need this fallback please add '--data-ciphers-fallback BF-CBC' to your configuration and/or add BF-CBC to -
-data-ciphers.
2026-02-07 18:39:03 Note: Kernel support for ovpn-dco missing, disabling data channel offload.
2026-02-07 18:39:03 OpenVPN 2.6.14 x86_64-pc-linux-gnu [SSL (OpenSSL)] [LZO] [LZ4] [EPOLL] [PKCS11] [MH/PKTINFO] [AEAD] [DCO]
2026-02-07 18:39:03 library versions: OpenSSL 3.5.0 8 Apr 2025, LZO 2.10
2026-02-07 18:39:03 DCO version: N/A
2026-02-07 18:39:03 net_route_v4_best_gw query: dst 0.0.0.0
2026-02-07 18:39:03 net_route_v4_best_gw result: via 192.168.170.2 dev eth0
2026-02-07 18:39:03 Diffie-Hellman initialized with 2048 bit key
2026-02-07 18:39:03 TUN/TAP device tun1 opened
2026-02-07 18:39:03 net_iface_mtu_set: mtu 1500 for tun1
2026-02-07 18:39:03 net_iface_up: set tun1 up
2026-02-07 18:39:03 net_addr_v4_add: 10.8.0.1/24 dev tun1
2026-02-07 18:39:03 Could not determine IPv4/IPv6 protocol. Using AF_INET
2026-02-07 18:39:03 Socket Buffers: R=[212992→212992] S=[212992→212992]
2026-02-07 18:39:03 TCP/UDP: Socket bind failed on local address [AF_INET][undef]:1194: Address already in use (errno=98)
2026-02-07 18:39:03 Exiting due to fatal error
2026-02-07 18:39:03 Closing TUN/TAP interface
2026-02-07 18:39:03 net_addr_v4_del: 10.8.0.1 dev tun1

┌──(root㉿kali)-[/etc/openvpn/easy-rsa]
└─#
```

relancement propre :

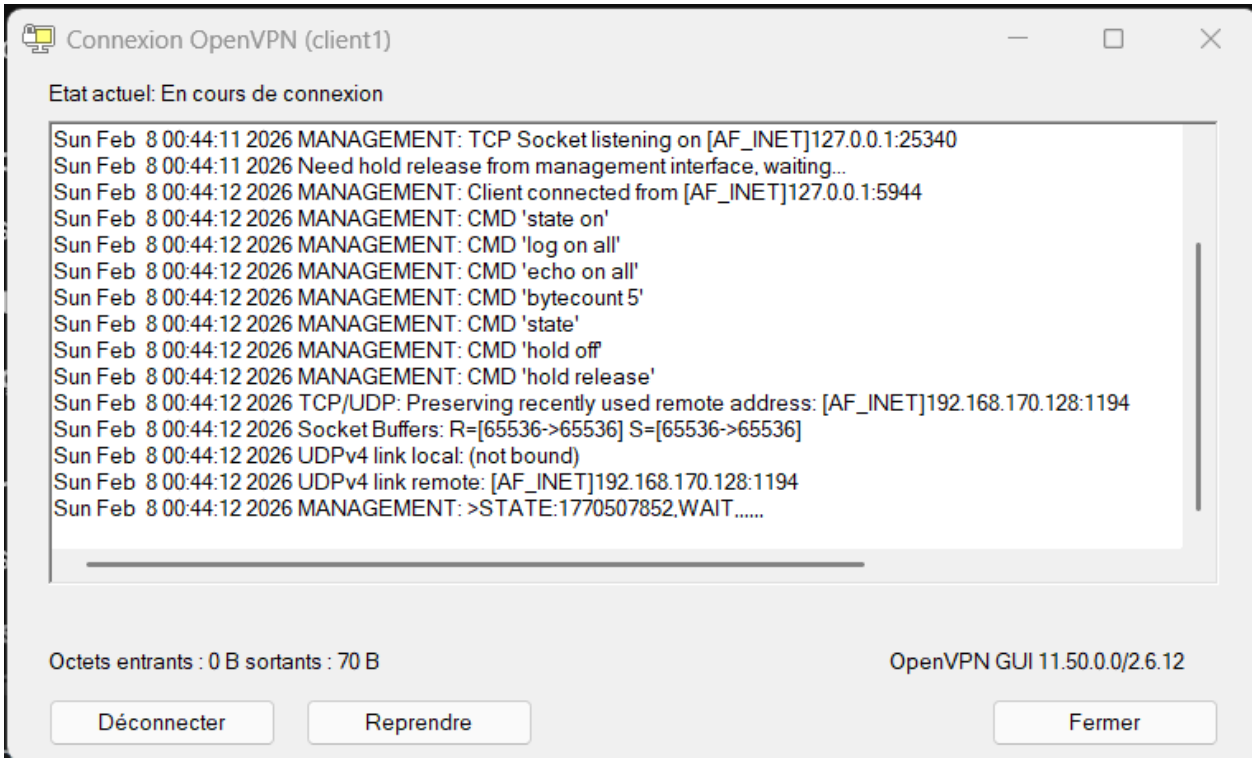
```
┌──(root㉿kali)-[/etc/openvpn/easy-rsa]
└─# killall openvpn

┌──(root㉿kali)-[/etc/openvpn/easy-rsa]
└─# systemctl restart openvpn@server

┌──(root㉿kali)-[/etc/openvpn/easy-rsa]
└─# ss -tuln | grep 1194
udp   UNCONN 0      0            0.0.0.0:1194      0.0.0.0:*

┌──(root㉿kali)-[/etc/openvpn/easy-rsa]
└─#
```

en attente du client




connecté depuis la machine hôte réussi !